

A Company of Legend: The Legacy of Fairchild Semiconductor

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Computer History Museum

Founded in 1957 by eight dissidents from Shockley Semiconductor Laboratory, Fairchild Semiconductor developed some of the most important innovations in 20th century technology and sowed the seeds of the microelectronic-driven computer industry of today. This article surveys some of the key personalities who worked at the company and describes their technological and management contributions.

Fairchild Semiconductor was a company of legend—perhaps the most extraordinary collection of business talent ever assembled in a start-up company. If Fairchild had a corporate culture it could only be described as volatility incarnate.... brilliant young engineers and marketers working long days, and partying long nights.... And somehow in the middle of it all, they also managed to invent the integrated circuit, the defining product of the late twentieth century, and in the process helped to create the modern world.¹

—Michael Malone (2007)

The general public likely remembers 1957 as the year Hank Aaron was named National League MVP player, “West Side Story” opened on Broadway, the literary antitheses *The Cat in the Hat* and *On the Road* were published, and the US Congress passed the Civil Rights Act. More technically inclined observers might recall the Soviet Union’s Sputnik launch, the resulting space race, and the renewed impetus it gave to the importance of science and technology in the US. Few are aware that 1957 also marked the founding of Fairchild Semiconductor, an upstart company that through an unprecedented series of technology, business, and cultural innovations, intended and otherwise, spawned Silicon Valley and changed the world.

This article was inspired by stories told, memories rekindled, and artifacts collected at “Fairchild@50,” a celebration of the 50th anniversary of the company’s founding sponsored by Fairchild alumni and held at Stanford University and the Computer History Museum in October 2007. It describes the background behind the founding of the

company and summarizes those areas of the semiconductor business where Fairchild established a significant market position or where technology pioneered by its engineers made an important contribution to the industry and the later success of spin-out and other competitors, including Advanced Micro Devices (AMD), Intel, and National Semiconductor.

Fairchild overview

Founded in September 1957 in Palo Alto, California, by eight young engineers and scientists from Shockley Semiconductor Laboratories, Fairchild Semiconductor pioneered new products and technologies together with an entrepreneurial style and manufacturing and marketing techniques that reshaped the semiconductor industry. The planar process—conceived just three months later and developed in early 1959—revolutionized semiconductor device production and continues to enable the manufacture of billion-transistor microprocessor and memory chips today.

Funded by and later acquired as a division of Fairchild Camera and Instrument Corporation of Syosset, New York, Fairchild was the first manufacturer to introduce silicon transistors matching the speed of germanium devices and practical monolithic integrated circuits (ICs) to the market. At the peak of its influence in the mid-1960s, the division was one of the world’s largest producers of silicon transistors and controlled more than 30% of the market for ICs. In 1965, Director of R&D Gordon Moore observed that device complexity was increasing at a consistent rate and predicted that this would continue

into the future. Moore's Law created a yardstick against which companies continue to measure their technology progress.

Starved by the parent company for funds needed to invest in new production facilities and retain key employees and torn by internal organizational issues, by the late 1960s the semiconductor division encountered serious problems with introducing new products and satisfying fast-growing customer demand. As Moore explained,

Fairchild grew to be about a \$150 million business and some 30,000 employees by the late sixties. It was a fairly significant corporation by the time we were done. But things began to deteriorate—partly, I think, because it was controlled by an East Coast company. . . . We had also made a tremendous number of mistakes, and we had squandered opportunities along the way. It was excellent on-the-job training, but there probably is a more efficient way of training entrepreneurs than by letting them make all the mistakes. Fortunately, good products make up for a lot of problems in an organization, and I think that was what happened in our case.²

In October 1967, a sharp industry downturn and loss of digital IC market share to the Texas Instruments (TI) transistor-transistor logic (TTL) family resulted in the division losing money for the first time since the start-up phase. In 1968, after all but one of the founders and many senior employees had left the company, a new team consisting of former Motorola executives led by C. Lester Hogan was appointed to head Fairchild Camera and Instrument. Hogan moved the corporate headquarters to Mountain View, and the company continued to innovate in new technologies and products, including the industry's first volume production of high-performance semiconductor memory devices and charge-coupled device (CCD) imagers that today form the sensors for digital cameras. Ventures into consumer products, including digital watches and videogames, were less successful. Although revenues grew substantially under the new regime, the company was never able to regain its former profitability and prominence.

Having been overtaken in sales and significance by companies started or managed by former Fairchild employees (or Fairchildren) and buffeted by other aggressive US and international competitors, the company was purchased as a diversification move by French

oilfield services conglomerate Schlumberger in 1979. Unable to restore its fortunes, Schlumberger sold the assets to National Semiconductor in 1987. In 1997, National divested a number of mature product lines in a leveraged buyout to a group of executives based at Fairchild's former South Portland, Maine, facility. Through internal development and strategic acquisition of compatible products for power, analog, logic, and mixed-signal applications, the reborn Fairchild Semiconductor is once again a public company with an annual revenue of more than \$1 billion.

Journalist Don C. Hoefler, who popularized the name Silicon Valley in 1971, published an article in 1968 that traced the lineage of 15 local semiconductor companies directly to Fairchild.³ Updated and printed by Semiconductor Equipment and Materials International (SEMI) as a wall chart in 1977, the number had increased to 66. A 1985 revision showed 125 spin-outs. Including systems and software businesses, the total number of companies in the Bay Area and beyond with Fairchild roots now lies in the thousands.

San Francisco Peninsula industrial boom

Commercial and amateur radio electronics activity began to emerge on the San Francisco peninsula more than 100 years ago. Surrounded on three sides by maritime waters, the area was an early developer of ship-to-shore radio technology. Stanford graduate Cyril Elwell made one of the first "mobile" wireless phone calls from downtown Palo Alto in 1908.⁴ While working for Elwell's Federal Telegraph Corporation, Lee de Forest built the first vacuum-tube amplifier and oscillator circuits using his three-electrode Audion device a couple of years later.⁵ Another Stanford alumnus Charles Herrold began the nation's first regularly scheduled music and entertainment broadcasting from San Jose's radio station FN in 1909.⁶

Several important component manufacturers emerged in the 1930s. Entrepreneur Charles Litton established Litton Engineering Laboratories to produce vacuum-tube-making equipment. William Eitel and Jack McCullough founded high-power tube maker Eitel-McCullough (Eimac). At Stanford the brothers Russell and Sigurd Varian, together with William Hansen and Edward Ginzton, founded Varian Associates and developed the first microwave tube light enough for airborne use, the klystron.⁷ After World War II, many other companies joined these pioneers in producing high-frequency electronic



Figure 1. The Fairchild founders pose under the Fairchild Semiconductor “Flying F” logo in the lobby of the Whisman Road headquarters building in Mountain View in 1960. From left to right, Gordon Moore, C. Sheldon Roberts, Eugene Kleiner, Robert Noyce, Victor Grinich, Julius Blank, Jean Hoerni, and Jay Last. (Courtesy of Wayne Miller/Magnum Photos)

components and systems for military and communications markets. They also established a supporting infrastructure of specialized suppliers as well as cadres of skilled operators and technicians. By 1955, regional employment in electronic-component-related activities reached about 3,000 workers.⁸

Fairchild emerges

In September 1955, William Shockley (coinventor of the transistor at Bell Laboratories) and industrialist Arnold Beckman founded the Shockley Semiconductor Laboratory as a division of Beckman Instruments “to engage promptly and vigorously in activities related to semiconductors.” Shockley rented a building at 391 South San Antonio Road in Mountain View and began recruiting “the most creative team in the world for developing and producing transistors.” He attracted a group of extremely capable young engineers and scientists who developed the basic equipment and processes for fabricating diffused silicon semiconductor devices. In December 1956, Shockley shared the Nobel Prize in

Physics for inventing the transistor, but by early 1957, his staff had become disenchanted with his abrasive management style. They also felt that the company should pursue near-term market opportunities for silicon transistors rather than the distant promise of the technically challenging four-layer (PNPN) diode for telephone-switching applications that Shockley championed.⁹

After unsuccessfully petitioning Beckman to appoint a new manager, eight of Shockley’s most talented employees—Julius Blank, Victor Grinich, Jean Hoerni, Eugene Kleiner, Jay Last, Gordon Moore, Robert Noyce, and Sheldon Roberts (see Figure 1)—resigned in September 1957 and founded Fairchild Semiconductor in nearby Palo Alto. Shockley continued pursuing his four-layer diode, but his company never realized a profit. Beckman sold the operation to Clevite Corporation in 1960, and Shockley became a professor of electrical engineering and applied science at Stanford University.

The founding of Fairchild stimulated an era of dramatic growth in the electronics

“The Legacy of Fairchild” in the Computer History Museum Collection

The Computer History Museum is home to one of the largest international collections of computing artifacts in the world. To some degree, almost every artifact based on semiconductor devices produced after the early 1960s depends in some way on technologies that were created by Fairchild or former Fairchild employees. The following are five of the most significant early machines in the collection that employ Fairchild devices:

- Autonetics Minuteman I Missile Guidance Computer (1962). A potential reliability problem with the mesa transistors stimulates Jean Hoerni’s invention of the planar process. (Accession Number X1502.97)
- Raytheon/MIT Apollo Guidance Computer (1965). The first production computer to commit to use monolithic integrated circuits (ICs). (Accession Number X37.81A)
- Control Data Corporation 6600 Supercomputer (1964). The first high-performance computer to use silicon transistors. (Accession Number X1385.97F)
- Cray Research Cray-1A Supercomputer (1976). This iconic machine was powered by Fairchild ECL logic and memory devices. (Accession Number X1553.98A)
- Burroughs Illiac IV Supercomputer (1975). The first machine to employ all-semiconductor main memory. (Accession Number 102716109)

industry on the San Francisco peninsula. Ten years later, at the peak of the company’s influence, local electronic component industry employment reached nearly 20,000 people. It exceeded 125,000 by 2003, not including the hundreds of thousands of employees in the computer hardware, software, and supporting industries that sprang from the silicon.¹⁰

A venture-banking industry

Arthur Rock, a young security analyst at the New York investment bank Hayden Stone, arranged financing for the new company. Rock and his boss Arthur Coyle, who specialized in developing financial services for emerging technology companies, approached more than 30 potential corporate investors before interesting Fairchild Camera and Instrument’s founder and chairman Sherman Fairchild. The bankers negotiated one of the West Coast’s first venture capital deals, whereby Fairchild loaned the founders \$1.38 million in exchange for the right to purchase the company if it prospered. It did prosper and the founders enjoyed a handsome return on their personal investments of \$500 each.¹¹ Rock’s success encouraged his move to California in 1961 to form a partnership with local investor Tommy Davis. Together they invested \$3 million in companies, such as pioneering scientific computing Scientific Data Systems and Teledyne; the latter firm grew into one of the most successful technology conglomerates in the history of American business. Davis’s and Rock’s fund returned \$100 million to its investors. After establishing his own firm, Arthur Rock & Co in San Francisco, Rock worked with Fairchild cofounders Moore and Noyce to launch Intel in 1968.

The financial rewards enjoyed by the Fairchild founders did not go unnoticed by later employees who had received only modest stock grants, if any. As the potential returns available in high-technology investments became more apparent, symbiotic relationships between entrepreneurs and bankers blossomed into the now legendary capital of venture capital lining Sand Hill Road in Menlo Park. This local availability of investors that understood technology combined with an aggressive, risk-taking culture established the San Francisco Bay Area, and Silicon Valley in particular, as one of the world’s most vibrant centers of new company formation and high-wage job creation.

Technological breakthroughs

Over the next 10 or so years, Fairchild’s engineers made countless important contributions to the industry. (See the “The Legacy of Fairchild” in the Computer History Museum Collection” sidebar for details on the contributions of Fairchild innovations.)

Commercial double-diffused silicon transistors

As a result of the Sputnik success, military contractors adopted the use of digital electronic techniques in avionics equipment. Recognizing that germanium transistors would not satisfy the extended operating temperature requirements of such systems, the Fairchild founders identified an opportunity for high-speed silicon devices. TI was in production with silicon transistors using a diffusion process for the base region and an alloy technique for the emitter. However, they did not meet the IBM Federal Systems Division exacting specifications for a transistor to drive

magnetic core memory in the B-70 military aircraft computer system.

Bell Labs had produced double-diffused (emitter and base) devices that offered higher speed, but they were not available outside the company. The Fairchild group felt they could satisfy the IBM application and create a viable commercial product by developing their own double-diffused process. However, according to Moore, “we were far from developing a commercial device.”⁷ At the Fairchild@50 event, Jay Last noted, “Bell Labs believed that photolithography was impractical but both Bob Noyce and I agreed that unless we could make it work, we did not have a company.” In just five months, Sheldon Roberts set up a crystal-growing operation; Last and Noyce developed photolithographic masking techniques using three matched 16-mm movie-camera lenses; Moore developed new diffusion methods and established techniques for making electrical contacts with aluminum; and Julius Blank, Victor Grinich, and Eugene Kleiner built manufacturing and test equipment. After successful delivery to IBM, the first commercial double-diffused transistor, the 2N696/697 NPN device, was introduced to great acclaim at the Wescon electronics component trade show in August 1958.

Autonetics selected the new Fairchild transistor for a guidance-and-control system on the Minuteman ballistic missile, the era’s largest defense program. In late 1958, however, a potential reliability problem put the new firm’s survival at stake. Tiny particles flaking off the inside of the metal package threatened to short across exposed junctions on the silicon’s surface. Swiss-born physicist and cofounder Jean Hoerni, who had developed the diffusion profiles for the company’s mesa-structure silicon transistors, proposed a solution that revolutionized the industry.

The planar process

Hoerni had recorded the essential ideas of a new process in his notebook on 1 December 1957 that entailed leaving the oxide masking layer in place on top of the silicon wafer to protect the sensitive junctions between the underlying p- and n-type semiconductor regions.¹² Due to concerns about possible contamination by the oxide, conventional wisdom required removing this layer after completing masking operations, thus exposing the junctions. Recalling his idea, Hoerni viewed the oxide instead as a possible solution to the exposed junction problem. His

planar approach, named after the finished device’s flat topography, would protect these junctions. After writing a patent disclosure in January 1959, he demonstrated a working planar transistor that March. As Hoerni had predicted, the oxide layer protected the junctions and was eventually qualified as the preferred Minuteman transistor process.

Planar devices also proved to have better electrical characteristics—particularly significantly lower leakage currents, which are critical in computer logic design. Fairchild introduced the 2N1613 planar NPN transistor commercially in April 1960. With a guaranteed useful current gain (beta) parameter over an operating current range of 100 μ A to 0.5 A, the device was received enthusiastically by the market. When matched with a PNP transistor counterpart, Fairchild products enjoyed a several-year monopoly in high-reliability applications that resulted in fast-growing sales and high margins. According to that year’s annual report, “By the end of 1961, Fairchild Semiconductor was the largest producer of high performance silicon transistors in the United States.”

Translating the planar concept into a high-volume production (see Figure 2) took extraordinary efforts in process control engineering and fabrication equipment design ingenuity, but production volumes quickly surpassed all previous techniques used for manufacturing semiconductors for both high-speed and low-power applications. Fairchild’s licensing of the planar process to other semiconductor companies encouraged standardized manufacturing processes across the industry. This in turn opened up opportunities for independent equipment producers and spawned an important new semiconductor equipment industry. Because it permitted fabrication of all the components of a semiconductor device from one side of a wafer, modern billion-transistor microprocessors and memories continue to rely on Hoerni’s breakthrough idea. Historian Christophe Lécuyer describes it as “the most important innovation in the history of the semiconductor industry.”⁷

Silicon transistors surpass germanium

In the early 1950s, Bell Labs’ scientists Howard Christensen and Gordon Teal developed a process called epitaxial deposition to grow a thin layer of material on a germanium wafer that continued the underlying crystalline structure. This offered the ability to

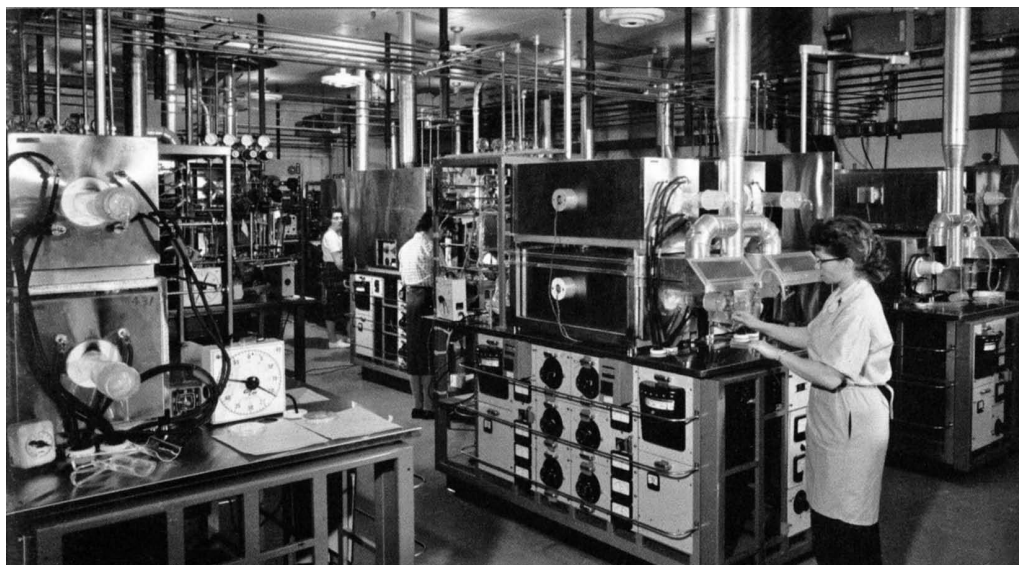


Figure 2. The Fairchild Semiconductor diffusion area in 1960. (The Computer History Museum Collection, courtesy of National Semiconductor)

form layers having different doping characteristics than the substrate. Henry Theurer of Bell Labs published the first description of a silicon transistor taking advantage of this process to improve speed and increase the breakdown voltage in 1960. Fairchild engineers were pursuing epitaxial technology in this same timeframe but lacked the equipment necessary to bring a device to market until March 1961.¹³ The resulting 2N914 became one of the company's most successful transistors. When later applied to the manufacture of ICs, epitaxy offered up to a 50× improvement in production yields.

Seymour Cray of Control Data Corporation awarded Fairchild a development contract for a silicon transistor that switched in less than 3 nanoseconds (ns). Existing silicon transistors offered operation superior to germanium at the elevated ambient temperatures prevalent in large computers but were too slow for Cray's planned new model 6600 supercomputer. Jean Hoerni met CDC's demanding specification by combining gold-doping—the addition of gold impurities (a technique he had developed for use on the company's original product line in 1959)—together with the new epitaxial deposition process. The 2N709 NPN device was introduced in July 1961 as the first silicon transistor to exceed germanium speed.¹⁴ In 1964, CDC placed "one of the largest single orders in the history of the semiconductor industry" with Fairchild for more than 10 million of the devices.

Monolithic ICs

As electronic systems grew more complex, engineers sought simpler ways to interconnect the thousands of transistors used to implement them. Government agencies funded micromodule and multichip hybrid circuit projects in search of a solution to this "tyranny of numbers." In 1952, Geoffrey W.A. Dummer of England's Royal Radar Establishment proposed, "With the advent of the transistor ... it seems now possible to envisage electronic equipment in a solid block."¹⁵ In 1953, Harwick Johnson filed a patent for a single-chip-phase shift oscillator. He also worked with Torkel Wallmark on an "integrated semiconductor device" concept announced by the Radio Corporation of America (RCA) in 1958.¹⁶ In October 1955, Arthur D'Asaro and Ian Ross of Bell Labs fabricated a four-stage counter for telephone applications. Yasuro Tarui of Japan's Ministry of International Trade and Industry (MITI) described a device in 1957 that integrated both junction and field-effect transistors (FET) on one chip.¹⁷ Shockley's ill-fated four-layer diode was an attempt to integrate the function of multiple separate components onto a single chip. While achieving various degrees of integration, none of these ideas yielded a general-purpose solution.

On 12 September 1958, TI's Jack Kilby built a circuit using germanium mesa PNP transistor slices he had etched to form transistor, capacitor, and resistor regions. Using fine gold flying wires, he stitched them

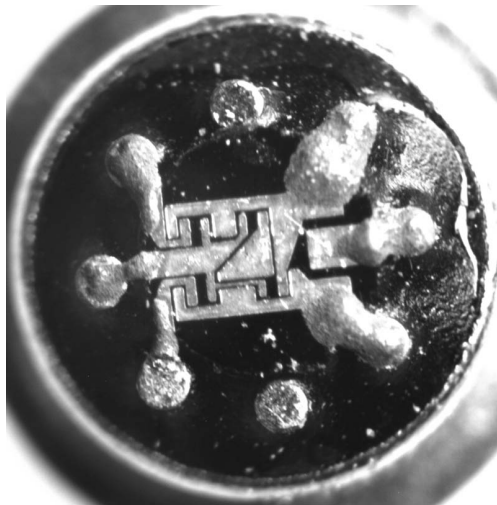


Figure 3. A mid-1960 implementation of the first Micrologic flip-flop using a physical isolation technique. (The Computer History Museum Collection, gift of Lionel Kattner)

together to create an oscillator function. One week later he demonstrated an amplifier. TI announced Kilby's concept in March 1959 and introduced the first commercial solid circuit, the Type 502 Binary Flip-Flop, at \$450 each in 1960. While the 502 and other members of TI's solid-circuit family demonstrated the concept of fabricating both active and passive components of an electronic circuit from semiconductor material, it did not address the critical manual interconnection challenges behind the tyranny-of-numbers problem.

Challenged by patent attorney John Ralls to identify other uses for Hoerni's planar process, Fairchild cofounder Noyce conceived the idea for a monolithic IC.¹⁸ The diodes, transistors, resistors, and capacitors diffused into the silicon wafer could be interconnected by depositing aluminum metal lines directly on top of the protective oxide coating rather than using Kilby's hand-stitched gold wires. Noyce filed his patent in July 1959.

Fairchild and TI engaged in litigation over IC patents for many years. The Supreme Court eventually ruled in Fairchild's favor, but by then, the companies had already settled on a cross-license agreement that included a net payment to Fairchild. Kilby and Noyce both received the National Medal of Science and today are celebrated as coinventors of the IC. Noyce died in 1990, but many believe he would have shared the Nobel Prize awarded to Kilby in 2000.

First planar IC

In August 1959, Noyce asked cofounder Jay Last to construct a concept microcircuit to "show the flag" at the Wescon component trade show. Last fabricated "a little device with four individual transistors in it and resistors from pencil lead" that they dubbed the Ticonderoga process, after the pencil company. In September, Last assembled a team including Sam Fok, Isy Haas, Lionel Kattner, and James Nall to develop a monolithic IC based on the Hoerni and Noyce patents. Robert Norman, head of the applications department under Vic Grinich, designed a modified direct-coupled transistor logic (DCTL) circuit, later called resistor transistor logic (RTL), as most compatible with the first generation of planar processing.¹⁹

Integrating multiple interconnected active and passive devices on one chip posed many new implementation challenges. The Noyce patent described a concept but did not define how to build a product. Over the next 12 months, Last's group developed many innovative approaches to accomplish the task. Their first working monolithic device with four transistors and five resistors in a flip-flop configuration was tested on 26 May 1960 (see Figure 3). Components were electrically isolated from each other by etching physical isolation channels into the silicon from the rear of the wafer and back-filling with epoxy. A more robust diffused p-n junction-isolation solution, requiring a difficult 20-hour boron diffusion devised by Isy Haas and Lionel Kattner with advice from diffusion expert David Allison, yielded working circuits on 27 September 1960. Under the trade name Micrologic, the type F flip-flop function was introduced to the public in March 1961 via a press conference at the Institute of Radio Engineers (IRE) Show in New York and a photograph in *Life* magazine. Six additional logic building blocks, including the type G (3-input NOR) gate function, a half adder (type H), and a half shift register (type S) were added to the family before the end of the year. In December 1961, an internal sales memo stated, "We are at a point now where we can discuss micrologic as a true product. Our shipments by the end of December will be in the \$500,000 region and the current backlog is about 5,000 units."

Using six different Micrologic part types, General Motors AC Spark Plug Division's MAGIC I system was "the first complete airborne computer to be mechanized

exclusively with integrated circuits” and was operational in late 1962. Against the advice of conservative advisers who warned that ICs might not be reliable, system design contractor Massachusetts Institute of Technology (MIT) Instrumentation Laboratory selected the Fairchild type G gate function as the basic logic-building block for NASA’s Apollo Guidance Computer (AGC). The design was upgraded to a dual-gate device in 1963. Built by Raytheon, each system used approximately 4,000 ICs. Consuming 200,000 units at \$20 to \$30 each, the AGC was the world’s largest user of ICs through 1965.²⁰

Early MOS technology

In 1959, M.M. (John) Atalla and Dawon Kahng at Bell Labs had achieved the first successful insulated-gate field-effect transistor (IGFET), which had been anticipated as early as 1926¹³ by overcoming the long troublesome surface states, electrons at the silicon surface that blocked electric fields from penetrating into the semiconductor material. Today the device is known as a metal-oxide-semiconductor (MOS) transistor after its sandwich structure. Although the MOS structure was much smaller than bipolar devices and it consumed less power, because of its lower speed, the device addressed no pressing needs in the telephone system and it was not pursued further at Bell Labs. In a memorandum written in 1961, however, Kahng pointed out its potential “ease of fabrication and the possibility of application in integrated circuits.”

Researchers at Fairchild Semiconductor and RCA, however, did recognize these advantages in 1960 when Jean Hoerni fabricated a MOS diode (Fairchild) and Karl Zaininger and Charles Mueller fabricated a MOS transistor (RCA). In early 1961, Fairchild’s C.T. Sah demonstrated a MOS tetrode device. Both companies introduced commercial products in 1964.

In a 1963 paper coauthored with C.T. Sah, Frank Wanlass of Fairchild’s Palo Alto R&D Laboratory showed that when in standby mode logic circuits combining a p-channel and an n-channel MOS transistor in a complementary-symmetry circuit configuration drew close to zero current.²¹ Wanlass patented the idea, now known as complementary MOS (CMOS), but manufacturing and reliability challenges prevented Fairchild and the industry from exploiting the technology’s low-power advantage in the near term. It took collaboration between

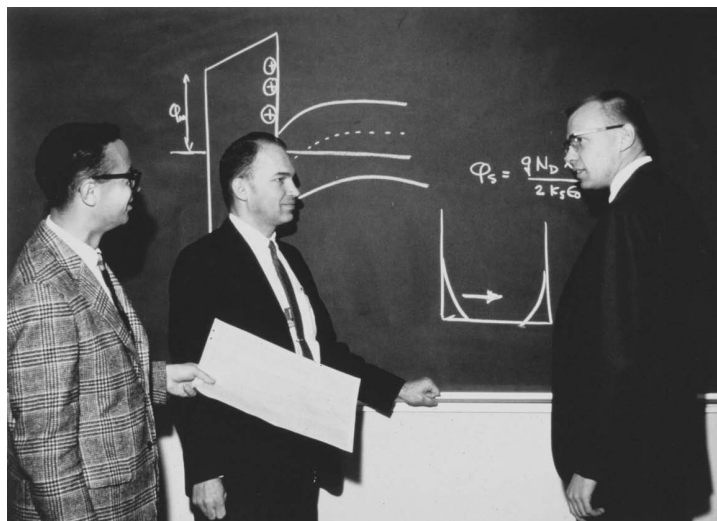


Figure 4. Andy Grove, Bruce Deal, and Ed Snow discuss metal-oxide-semiconductor (MOS) technology at Fairchild R&D in 1966. (The Computer History Museum Collection, courtesy of National Semiconductor)

researchers from industry and academia around the globe to resolve the fundamental issues that allowed CMOS to emerge as the dominant IC technology of today. Important contributions were made by the pioneering teams at Fairchild and RCA. Between 1963 and 1966, Bruce Deal, Andrew Grove, and Ed Snow at Fairchild identified the significant issue of sodium contamination and published numerous papers that explained the electrical nature of oxides (see Figure 4). Technologist Federico Faggin believes that “Fairchild developed the most cogent understanding of the silicon-silicon dioxide interface of the late 1960s.” Much of this work informed Grove’s classic textbook *Physics and Technology of Semiconductor Devices*, published in 1967.

Fairchild was also an important contributor to the next major improvement in speed and density of MOS devices, the application of silicon-gate technology to ICs. Bell Labs had showed that the speed, reliability, and packing density of discrete MOS transistors could be improved by replacing the conventional aluminum metal gate electrode with a polycrystalline layer of silicon. At Fairchild R&D, Faggin and Tom Klein adapted the process to ICs. Faggin redesigned an existing metal-gate, eight-channel analog multiplexer to take advantage of the self-aligning gate feature, and in 1968, Fairchild introduced the first commercial silicon-gate IC, the 3708. Faggin’s other innovations developed at Fairchild—the buried contact and the bootstrap load—played a critical role in

implementing Intel's first microprocessor (the 4004) after he joined the latter company in 1970.²²

Standard logic families

With its early-to-market advantage, Fairchild Micrologic was the most popular first-generation standard digital IC family. Signetics introduced an easier to use alternative in 1962, the first commercial integrated diode transistor logic (DTL) family. In a competitive industry leap-frog pattern that became characteristic of the business, Fairchild's improved 930 Series DTL, designed by Dick Bohn and Robert Seeds in 1964, quickly gained acceptance and dominated the market through much of the 1960s.²³

Inspired by an application note on "New Forms of All Transistor Logic," written by Fairchild's H. Ruegg and R. Beeson in 1961, Thomas Longo developed the first commercially successful TTL family, Sylvania Universal High-Level Logic (SUHL) in 1963.²⁴ TI implemented a lower-cost SN7400 series TTL in 1966 and, through aggressive marketing and high-volume manufacturing capability, surpassed Fairchild as the leading logic supplier by the end of the decade. Through technological initiatives, such as defining the most popular TTL medium-scale integration (MSI) functions, the 9300 Series, and applying Isoplanar oxide-isolation technology to deliver the highest-speed family with Fairchild Advanced Schottky Technology (FAST), the company held second place in digital bipolar logic sales behind TI during much of the 1970s. Fairchild also developed a very-high-speed emitter coupled logic (ECL) family, designated 100K, which enabled the Cray 1 supercomputer in 1976. However, the company never regained its initial market prominence.

Linear IC market breakthrough

Analog (or linear) circuits amplify and condition signals from continually varying phenomena, such as sound, temperature, and radio waves. Because of the nearly infinite resolution required to process these signals, analog circuits demand high precision in design and manufacturing. Fairchild as well as Amelco, Plessey, RCA, TI, Westinghouse, and others developed integrated versions of simple differential amplifiers as early as 1962, but they achieved limited market acceptance.

It took the unique genius of the colorful young designer Robert Widlar working with creative process engineer Dave Talbert to conceive the first widely used linear device, the Fairchild μ A702 operational amplifier (op-amp) in 1964. Believing that a traditional sales approach would not adequately promote his product, Widlar demanded and received possibly the industry's first dedicated linear-circuit marketing manager.²⁵ Jack Gifford moved up from the Los Angeles sales office to begin a long and successful career in the analog IC business. He retired as chairman of Maxim Integrated Products in 2007.

Widlar and Talbert's successor device, the μ A709 op-amp, solved important design quirks of the μ A702 to establish a mass market for analog ICs and a highly profitable business unit for Fairchild. In late 1965, the pair departed for Molex (later acquired by National Semiconductor), where they built a linear dynasty that eventually surpassed Fairchild's. In another example of technological competitive leap frog, by adding an internal compensating capacitor to one of Widlar's early designs for National, the LM101, Fairchild R&D engineer Dave Fullagar created the μ A741, the most popular op-amp of all time.²⁶ Long after his retirement, Fullagar's circuit remains in production today.

Digital is the most efficient form for manipulating many kinds of information. However, real-world data is analog and must be converted to digital form for processing. ICs incorporating analog and digital circuitry where signals are translated between these two modes are called mixed-signal devices. In 1968, Fairchild designer George Erdi developed the μ A722 10-bit Current Source, one of the first IC building blocks specifically created to integrate an important functional component of many data-conversion circuits. With the proliferation of digital consumer products, mixed-signal devices represent one of the fastest-growing sectors of the industry today.

Dual-inline package

Semiconductor devices are typically enclosed in a robust outer package to prevent damage to the chip and its fragile connecting wires. Despite this important consideration, packaging was one of the most neglected aspects of early IC design. Typically, an existing transistor package was modified to squeeze in an IC chip.

The first Micrologic devices were etched into a disk shape small enough to fit inside the pin-circle of a TO-18 can. (TO stands for transistor outline.) None of these formats were convenient for a high-density, printed circuit board layout.

In 1965, Fairchild's Don Forbes, Rex Rice, and Bryant ("Buck") Rogers devised a 14-lead ceramic dual-inline package (DIP) with two rows of pins 100 mils apart that revolutionized computer manufacturing by simplifying layout and allowing automated insertion into printed circuit boards.¹⁰ Versions of the DIP with pin counts increased up to 64 leads and molded with low-cost, plastic compounds dominated production volumes until surface mount technology became widely used in the 1980s.

Moore's self-fulfilling prophecy

In 1965, Fairchild Semiconductor's Director of R&D Moore published the article "Cramming More Components onto Integrated Circuits" in *Electronics* magazine. Moore's goal was to encourage computer designers to take advantage of future, higher-density ICs by predicting "the development of integrated electronics for perhaps the next ten years." He drew a line through five data points representing the number of components per IC for minimum cost per component developed by Fairchild between 1959 and 1964. Extrapolating the trend to 1975, he projected that the practical number of components per chip would reach 65,000, doubling every 12 months.²⁷

At the 1975 IEEE International Electron Devices Meeting, Moore (then Intel's president and CEO) noted that advances in photolithography, wafer size, process technology, and "circuit and device cleverness," especially in semiconductor memory arrays, had allowed his 1965 projection to be realized. Adding more recent data, that included a higher mix of microprocessor designs that were somewhat less dense than the memory chips that had sustained his initial trend line, he slowed the future rate of increase in complexity to "a doubling every two years, rather than every year." This prediction became a self-fulfilling prophecy that emerged as one of the semiconductor industry's driving principles. Technologists worldwide were challenged, and continue to be challenged, with delivering regular breakthroughs that ensure compliance with Moore's Law, as it was later dubbed by Carver Mead.²⁸

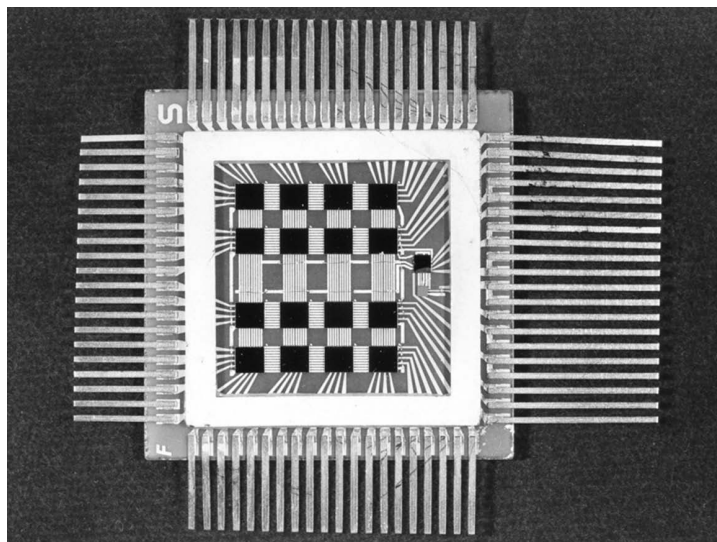


Figure 5. A semiconductor active memory (SAM) multichip 1,024-bit metal-oxide-semiconductor (MOS) memory plane from 1968. (The Computer History Museum Collection, courtesy of National Semiconductor)

Pioneering semiconductor memory activities

According to Moore, in the early days of the Micrologic family development, Bob Norman suggested that multiple semiconductor flip-flops could be used to build a memory array but he "decided it was so economically ridiculous, it didn't make any sense to file a patent on it."²⁹ However, by 1963, Moore had changed his opinion and described the possibility of semiconductor memory as "especially inviting." John Schmidt designed a 64-bit MOS p-channel static random access memory (RAM) in 1964, and in 1965, a Memory Products business unit was established at 2525 Charleston Road, Mountain View. A 13-member R&D staff was chartered with developing both ferrite core and monolithic semiconductor memory assemblies. The company employee journal *Leadwire* (October–December 1965) noted that "semiconductor IC memory systems pioneered by Memory Products promise to be the wave of the future."

One early result of this effort was a series of multichip memory modules described as semiconductor active memory (SAM) planes (see Figure 5). In 1968, Zeev Droori, who later founded Monolithic Memories (MMI), managed the development of a 1,024-bit SAM module for Burroughs comprising 16 of Schmidt's 64-bit RAMs face-down bonded to a 1.5 × 1.5 inch ceramic substrate that

formed the base of an 80-lead package (see Figure 5). Multiple planes were then stacked three inches high to form complete memory systems of up to 16,000 bits. However, monolithic solutions soon overtook this multichip approach.

Fairchild applications engineer Lee Boysel promoted the use of a dynamic four-phase circuit technique to increase the speed and packing density of MOS circuits. In 1967, he defined a family of computer-oriented MOS large-scale integration (LSI) building blocks. Several of these functions, including a 256-bit dynamic RAM (DRAM), were fabricated before he left to found Four Phase Systems in 1968. At Four Phase, he used this work as the basis for a 1,024-bit DRAM used in the company's System IV/70 computer. The device preceded merchant market DRAMs by several years.¹⁶

Fairchild was never able to leverage these important pioneering efforts in MOS semiconductor memory into a significant market position, but it did become a dominant player in high-speed bipolar memory devices.

A 1966 Fairchild Semiconductor IC short-form catalog included four bipolar memory devices of up to 16-bit density offered in current mode logic (CML), DTL, and TTL compatible versions. They were designed for scratch-pad memories and for local and bulk storage "to allow the computer manufacturer to realize fully integrated systems at the lowest possible cost." At the 64-bit density level, Fairchild competed with concurrent offerings from Sylvania and TI but was first to market with the next generation 256-bit TTL device. Designed by H.T. Chua before he left to craft Intel's first memory chip and later MMI's programmable-array logic (PAL) device, at 104×134 mils, the 4100 (aka 93400) was one of the largest production chips of its era.

The Illiac IV was a high-performance computer system funded by the US Department of Defense's Advanced Research Project Agency (ARPA). The project was conceived and managed by Daniel Slotnick of the University of Illinois, and the system hardware was designed and built by Burroughs. The Illiac IV initial specifications called for a 2,048-word, 64-bits-per-word of memory and a total of 131,072 bits for each of the machine's 64 parallel processing elements. The memory was to operate with a cycle time of 240 ns and access time of 120 ns. When the planned thin-film memory could not deliver this performance, Burroughs

contracted with Rex Rice, manager of the Fairchild R&D Digital Systems Department, to build the systems using semiconductor devices in 1970. By May 1971, a team led by Frank Greene and Wendell Sander designed and shipped 70 processor element memory (PEM) systems to Burroughs using Chua's 256-bit TTL device. According to Slotnick, "Illiac IV was the first machine to have all-semiconductor memories. Fairchild did a magnificent job of pulling our chestnuts out of the fire ... the memories were superb and their reliability to this day is just incredibly good."³⁰

In 1971 Fairchild announced a 256-bit TTL RAM (93410), the first commercially available memory using an oxide-isolated bipolar process that the company called Isoplanar.³¹ Developed by Douglas Peltzer, the technology was conceptually similar to Jay Last's physical isolation approach used to make the first planar ICs, the epoxy-filled trenches being replaced by thick oxide walls. Isoplanar's smaller die sizes and improved performance enabled Fairchild to establish the industry's largest and most profitable memory business unit producing TTL and ECL bipolar chips for high-speed applications. Seymour Cray continued his earlier working relationship with the company when he designed the iconic 1976 Cray 1 supercomputer, using up to 65,000 Fairchild 1024-bit ECL RAM chips per system together with 250,000 dual-ECL gate elements.

Beginnings of EDA

As ICs began to incorporate hundreds of gates and thousands of transistors, the computers they enabled were harnessed to speed the development process and eliminate design errors. IBM pioneered some of the earliest work in computer-aided design (CAD) and later electronic design automation (EDA). By 1966, James Koford and his colleagues at IBM Fishkill were capturing solid logic technology (SLT) hybrid circuit module designs for the Series 360 machines on graphical displays, checking them for errors, and automatically converting the information into mask patterns.

Recruited to Fairchild R&D by Moore and Bob Seeds, Koford worked with Hugh Mays, Ed Jones, Rob Walker, and others to apply this process to monolithic ICs. They designed graphic displays optimized for efficiently laying out mask designs, wrote logic simulators (FAIRSIM), and created test program

generators to work with Fairchild Instrumentation Division-designed computer-controlled test systems.³² Place-and-route software for implementing custom mask patterns laid the ground work for generations of electronic design automation tools and the beginnings of an important independent EDA industry in the Silicon Valley.

Gate arrays were developed by several vendors in the mid-1960s to speed the design turnaround time for custom ICs. The custom metal layer masks were designed manually. The Micromatrix family of gate arrays employed Fairchild CAD tools to perform this operation interactively. The Micromatrix 4500 array introduced in 1967 offered 32 DTL gates interconnected with two layers of metal. One year later, the 4700 array offered 150 TTL-compatible gates. Micromosaic standard cell products implemented customer designs in less silicon area but required custom masks for all production layers. Fairchild's first commercial MOS Micromosaic units sold to GE's Avionics unit in 1968. These ground-breaking programs were technically successful, but the high cost of computer time on the IBM 360/67 mainframe-powered CAD systems rendered the business uneconomical. Fairchild exited the business in the early 1970s.

In the 1980s, two companies founded by former Fairchild personnel associated with these programs established leading positions in the application-specific integrated circuit (ASIC) business by building on the pioneering efforts of their former employer. VLSI Technology was founded in 1979 by Jack Balletto, Dan Floyd, and Gunnar Wetlesen—all from Fairchild by way of Synertek—and Doug Fairbairn from Xerox Palo Alto Research Center (PARC) to offer standard cell devices supported by a broad suite of EDA tools. LSI Logic was founded in Santa Clara, California, in 1981 by former Fairchild CEO Wilfred Corrigan together with Mick Bohn, Jim Koford, Bill O'Meara, and Rob Walker to apply CAD software tools and fast turnaround prototype capability to deliver complex gate-array devices.

Further spin-out innovations

In Fairchild's continuous search for new product opportunities, Moore recalls "calling the senior people in the laboratory together and saying 'OK, we've done integrated circuits. What'll we do next?' And we started looking for all the peculiar physical effects we could find to see what new devices we

The success of Fairchild's Hong Kong plant began the high-tech industry drive to major offshore operations throughout Asia.

could invent."² Some of these ideas used compound semiconductor materials such as gallium arsenide and led to the establishment of a Microwave Products group in 1965. Under the leadership of former Bell Labs' MOS pioneer John Atalla, the business expanded in 1971 to embrace optoelectronic, radio frequency, and transducer products as the Microwave and Optoelectronics Division (MOD) on Deer Creek Drive in Palo Alto.

Gene Weckler, a member of the technical staff at the R&D Lab, worked on the development of processes and structures to improve silicon photodetectors and wrote a 1967 paper generally regarded as the original CMOS sensor paper. He came up with a design that resulted in a 128-element self-scanned linear sensor array product introduced in 1970. Together with MOS technologist Ed Snow, he founded Reticon in 1971 to commercialize these ideas.

The first planar technology gallium arsenide device, an infrared emitting diode, was produced in 1969 and followed the next year by monolithic, visible light-emitting diode (LED) display digits based on gallium arsenide phosphide. Along with many other Silicon Valley companies, Fairchild rode the boom in seven-segment LED displays for calculators and watches in the early 1970s until they were displaced by low-power, liquid crystal displays.

Continuing his early Bell Labs work on charge-coupled devices, Gil Amelio led Fairchild's design and manufacturing of CCD products for serial storage and imaging applications. The latter proved to have the most lasting value as the basis for today's scanner and digital camera sensors. The company announced the "world's first commercially available CCD device," a 1×500 element linear image sensor, in 1973.³³ Sold by Schlumberger in 1982, the business continues today in Milpitas as Fairchild Imaging.

Offshore assembly pioneer

Most early high-tech companies tended to expand within their immediate geographical region, but even as it committed to doubling its just-completed new plant size in Mountain View in 1959, Fairchild began to look outside the local area for further growth. The first facility beyond the San Francisco peninsula, a dedicated diode manufacturing plant in San Rafael, California, opened in 1960; the first outside the state was a test-and-assembly operation in South Portland, Maine, in 1962.

In 1963, Manager of Administration & Planning Jerry Levine launched a feasibility study on how to take advantage of the highly skilled, low-cost labor pool in Asia. Following a vacation in Hong Kong, he made a recommendation to Director of Manufacturing Charlie Sporck to open an assembly plant there. After starting in rented facilities, the board approved the construction of a company-owned high-rise building in 1964. Led by plant manager C.E. (Ed) Pausa, within a year Fairchild was the largest electronics employer in Hong Kong. Two years later there were 5,000 employees there and 3,000 in California. The success of this venture attracted the attention of other US component manufacturing companies and began the high-tech industry drive to major offshore operations throughout Asia.

A less-enduring assembly operation initiated in 1965 in cooperation with the Navajo Nation led to the construction of a 120,000 square foot facility employing 950 people on the Navajo Reservation at Shiprock, New Mexico. After resolving early cultural differences, the operation ran successfully and received several awards for its innovative practices. Internal political differences within the tribe resulted in an armed takeover of the plant and its subsequent closing in 1975.

Marketing and sales initiatives

Fairchild marketing mavens, beginning with the first head of sales and marketing Tom Bay and including such luminaries as Don Valentine (who later moved to National Semiconductor and Sequoia Capital), W.J. Sanders III (AMD), Robert Graham (Intel and Novellus), and Floyd Kvamme (National Semiconductor and Apple Computer), initiated practices that are now taken for granted in the sales and marketing of high-technology products worldwide. A systems and applications group published applications notes and designed printed

circuit boards showing how the new planar devices could be used most efficiently in end-product designs. Transistor radio designs based on Fairchild circuits proliferated throughout Hong Kong and other Asian countries. These are commonly known today as reference designs. Notably, applications engineers built a prototype set showing how high-frequency transistors could be used in color television.⁸ The concept was mass produced with only minor changes by GE, Sylvania, and Zenith using kits of parts supplied by Fairchild.

Creative approaches were developed to aid the sales force in getting the Fairchild story across to engineering decision makers. A flexible vinyl phonograph record with engineer Richard Anderson describing the characteristics and applications of the new Micrologic IC family was pressed and bound into the July 1961 issue of *EDN (Electrical Design News)* magazine. In an attempt to reach the broadest possible audience of potential customers on 10 October 1967, the company purchased air time on 32 TV stations that broadcast to the major centers of electronic systems design across the country. Fairchild's Harry Sello and Stanford Electrical Engineering professor Jim Angell presented a half-hour "Briefing on Integrated Circuits" in what is claimed to be one of the first TV infomercials. They described the steps in the fabrication of the planar process and the economic and competitive advantages of designing with ICs. Bold and extravagant use of the latest communications medium to promote high-tech products is common today, but it was unique and controversial in the 1960s.

Fairchild also signed Faust/Day, a new agency in Southern Californian, to develop aggressive advertising programs. The principals implemented an unusual campaign for a technology client by developing a brand image that included contracting the company name in the logo to "FAIRCH." One memorable ad entailed packing a low-cost line of plastic-encapsulated transistors in baked-bean cans and pricing them by the pound. According to advertising manager Elliott Sopkin, "No one had tried to sell technology like soap before." The agency merged into Chiat/Day in 1968 and went on to produce distinctive work for Apple Computer, including the Macintosh "1984" Super Bowl commercial.

Various organizational structures were tried and tested to improve the effectiveness

of the marketing and sales activities in winning designs and booking orders. One of the most lasting is a matrix of technically trained generalist sales personnel focused on major markets—such as computer, consumer, industrial, and military customers—supported by factory marketing functions aligned by in-depth product expertise. Not all these experiments were successful, but many of them resulted in methods and procedures that continue to inform the current generation of Silicon Valley Internet-focused marketeers.

Fairchildren and Silicon Valley culture

Although journalist Michael Malone's characterization of Fairchild's corporate culture as "volatility incarnate" might represent hyperbole, the management style and culture that emerged from Fairchild certainly differed from the straight-laced, bureaucratic, almost feudal ethos of its East Coast parent. Youth, inexperience, passion, a diverse mix of immigrants from across Europe and Asia, a tolerance for risk taking, and a strong engineering discipline evolved into a formula that was replicated across the valley as ex-employees spun-out to successor companies such as AMD, Intel, and National Semiconductor—and on to Apple, Atari, Netscape, and Sun as well as Cisco, Google, and Yahoo.

A recent magazine article proclaimed enthusiastically, "Fairchild's 50th anniversary commemorated not just the founding of a company but an epochal turning point in human evolution."³⁴ Since numerous books, academic treatise, and popular journals have expounded on this topic and the company's impact on the creativity and rise of Silicon Valley at great length, I will not explore it further here.

Fairchild cannot be assigned full credit or blame for the Silicon Valley phenomenon. However, there's little disagreement that the events that began with the spin-off of Fairchild from Shockley Semiconductor Labs in 1957 set the juggernaut in motion. Urban cultural historian Peter Hall has compared the Silicon Valley's rise to that of other great centers of creativity, including Florence during the Renaissance and Manchester, England, during the Industrial Revolution: "Creative cities . . . are places of great social and intellectual turbulence, not comfortable places at all."³⁵ Many of the Fairchildren who attended the October 2007 event described their tenure at the company as the most

rewarding professional experience of their lives. Few chose the adjective "comfortable."

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